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General Comments : Consolidated HW02 submission document uniting the UP Inception Overview (Vision, Personas, Requirements, Business Rules, Use Cases, Glossary) and the Agile Scrum Product Backlog, Themes/Epics, User Stories, Tasks, Sprint Plan, and PSI/MVP definition for the CSE305 Library Management System.

HW02 — UP Inception Overview & Agile Scrum Product Backlog

Library Management System (LMS)

Course	CSE305 Software Engineering — Adnan Menderes University (ADU), Department of Computer Engineering
Instructor	Assoc. Prof. Dr. Fatih Soygazi
Project	Library Management System (LMS)
Team & Scrum roles	**Ibrahim Temur** (`231805033`) — Product Owner *(proposed)* · **Omer Faruk Kuş** (`231805026`) — Scrum Master *(proposed)* (both: Developer, Architect, QA, Reviewer)
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Archive	`cse305_hw02_231805033_231805026.zip` (created with `python3 tools/package.py 02`)
GitHub repository	`https://github.com/ibrahimtemur/cse305-library-management-system`

> **Note**
> **How to read this document.** Part 1 provides a self-contained executive consolidation of the Unified Process (UP) Inception artifacts and links to the authoritative detail files in `docs/02-inception/` : `docs/02-inception/VISION_AND_CONTEXT.md` , `docs/02-inception/PERSONAS.md` , `docs/02-inception/REQUIREMENTS.md` (`FR-001` .. `FR-032` , `NFR-001` .. `NFR-012` , `CON-001` .. `CON-006` , `DR-001` .. `DR-007`), `docs/02-inception/BUSINESS_RULES.md` (`BR-001` .. `BR-018`), `docs/02-inception/USE_CASES.md` (`UC-01` .. `UC-11`), and `docs/02-inception/DOMAIN_GLOSSARY.md` . Part 2 presents the complete Agile Scrum deliverables: Themes & Epics (`docs/02-inception/THEMES_EPICS.md`), User Stories (`docs/02-inception/USER_STORIES.md` , `LMS-001` .. `LMS-021`), Ordered Product Backlog (`docs/02-inception/PRODUCT_BACKLOG.md`), Three-Sprint Iteration Plan (`docs/02-inception/SPRINT_PLAN.md`), Task Breakdown (`docs/02-inception/TASKS.md` , `105` tasks), and Potentially Shippable Increment / MVP Definition (`docs/02-inception/PSI_AND_MVP.md`). Every story point, classic hour, human hour, capacity calculation, commitment line, and traceability check is verified deterministically by `tools/agile/validate_backlog.py` .

Contents: Part 1 — UP Inception Overview · Part 2 — Agile Scrum Product Backlog & Iteration Plan · Revision History

Part 1 — UP Inception Overview

1.1 Vision, Problem Recap, Goals & System Context

Full specification: `docs/02-inception/VISION_AND_CONTEXT.md` .

Product Vision Statement

For university, faculty, and departmental libraries that manage physical book collections for academic staff, administrative staff, and students, **who** struggle with error-prone manual ledgers, untracked physical book copies, inconsistent overdue penalty calculations, and lack of real-time shelf visibility, **the Library Management System (LMS)** is a self-contained, web-based library catalog and circulation management application **that** automates bibliographic cataloging, barcode-level physical copy lifecycle tracking (`AVAILABLE` , `LOANED` , `RESERVED` , `MAINTENANCE` , `LOST`), policy-driven lending and renewal rules, First-In-First-Out (FIFO) title reservation queues, and deterministic overdue fine calculation. **Unlike** informal spreadsheet tracking or heavyweight commercial integrated library systems requiring external institutional infrastructure, **our product** operates 100% offline out-of-the-box with an embedded H2 database seeded from public-domain Open Library metadata (`782` books, `1,706` physical copies) and strictly synthetic operational data (`60` members, `3` librarians, `164` loans, `47` fines, `22` reservations), enforcing the confirmed Adnan Menderes University (ADU) library lending rules through modular GoF/GRASP design patterns and a JUnit 5 Test-Driven Development (TDD) test suite.

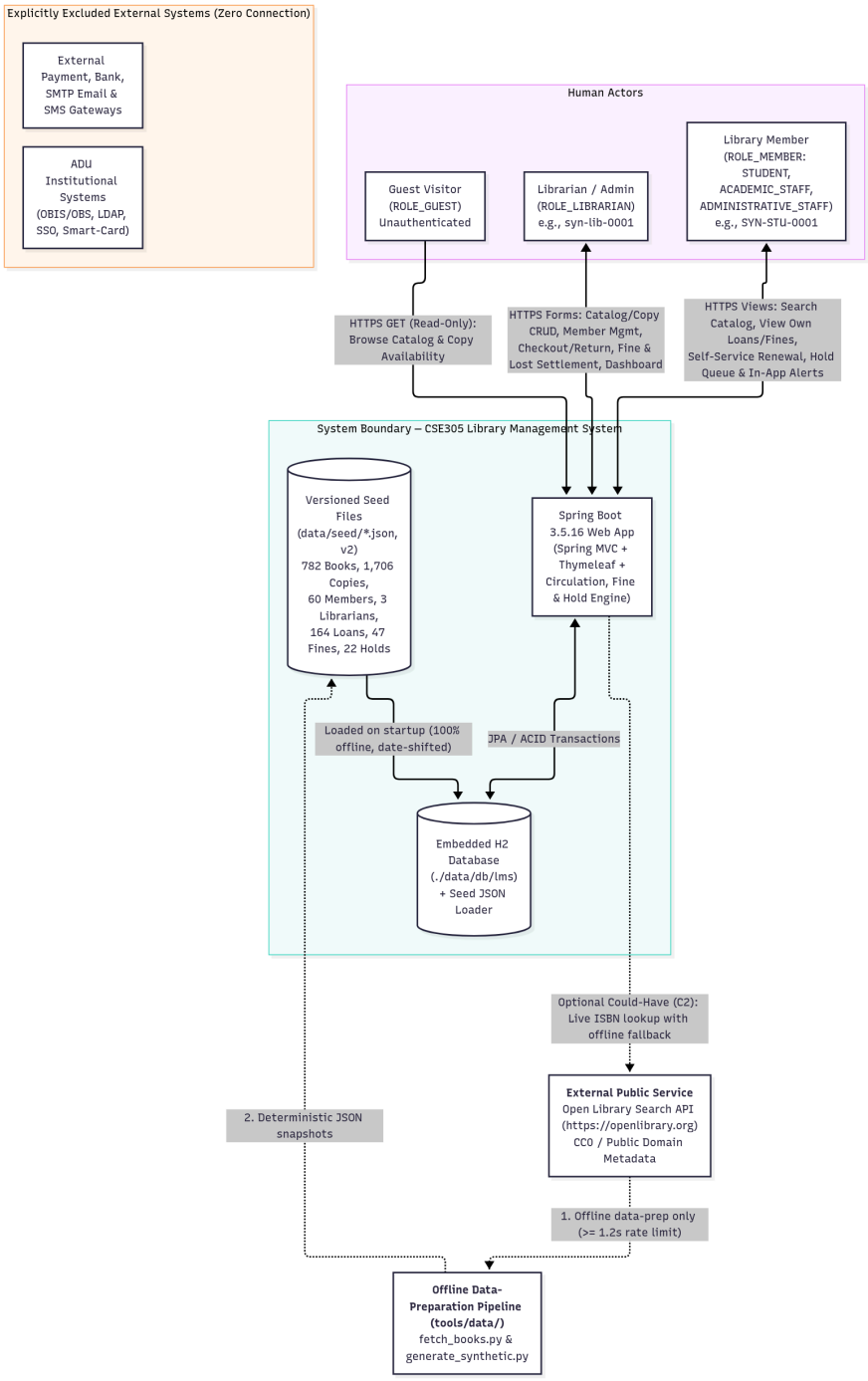
Problem Recap & Project Goals (`GOAL-01` .. `GOAL-05`)

Small-to-medium academic libraries using paper logs or unvalidated spreadsheets face four bottlenecks: (1) clerical overhead and manual date/ID errors at checkout/return, (2) inability to distinguish shelf-available copies from loaned, reserved, maintenance, or lost copies (or non-circulating reference works), (3) inconsistent enforcement of member quotas (`7` / `3` / `3`), `0.25 TRY/day` fines, post-renewal cooldowns, and borrowing blocks, and (4) zero self-service visibility for members.

Goal / Metric ID	Category	Summary Statement & Verification Basis
`GOAL-01` / `MET-01`	**Operational Automation & Policy Accuracy**	Replace manual ledgers with transactional checkout, return, renewal, reservation, and fine settlement governed by `BR-001` .. `BR-018` . **Target (`ASSUMPTION`):** `100%` pass rate across JUnit 5 tests covering quotas (`7` / `3` / `3`), `30` -day periods, `0.25 TRY/day` uncapped fines, `30` -day post-renewal cooldown, and blocks.
`GOAL-02` / `MET-02`	**Copy-Level Inventory & State Integrity**	Track every physical `BookCopy` (`LMS-COPY-1001` ..) via `BookCopyState` and block checkout of `referenceOnly` books (`BR-014` , `BR-017`). **Target (`ASSUMPTION`):** `0` illegal transitions or double-loaned copies; `100%` 1-to-1 invariant between `LOANED` copies and `ACTIVE` loans, and between `RESERVED` copies and `READY` holds.
`GOAL-03` / `MET-03`	**Self-Service & Local Query Latency**	Empower members to search the catalog, inspect loans/fines/cooldowns, renew, and place/cancel holds (`UC-01` , `UC-07` , `UC-08` , `UC-10`). **Target (`ASSUMPTION`):** `< 200 ms` local server response time on the reference workstation against the seeded H2 dataset (`NFR-005`).
`GOAL-04` / `MET-04`	**CSE305 Engineering Rigor & Startup Speed**	Apply UP modeling, Agile Scrum, UI-First prototyping via **Stitch MCP** (`NFR-001`), GoF/GRASP patterns (`NFR-010`), and TDD (`NFR-011`). **Target (`ASSUMPTION`):** `< 10 seconds` offline startup and seed initialization time with zero runtime internet calls (`NFR-004` , `NFR-006`).
`GOAL-05` / `MET-05`	**Offline Reproducibility & Synthetic Privacy**	`0` real personal names, student IDs, emails, or plaintext secrets in Git history; `100%` of seeded emails use `@synthetic.lms.local` and demo passwords are BCrypt-hashed at startup (`NFR-008` , `NFR-009` , `BR-018`).

System Scope & Explicit Exclusions

- **In Scope:** Printed books only (including non-circulating `referenceOnly` encyclopedias, dictionaries, and catalogs, `BR-014`); copy-level inventory (`1,706` seeded copies); synthetic `STUDENT` , `ACADEMIC_STAFF` , and `ADMINISTRATIVE_STAFF` members; desk checkout/return, self-service/desk renewal, `0.25 TRY/day` uncapped fine calculation, desk payment/waiver, lost-copy petition (`lostReportedOn` , fine freeze, 30-day replacement window, `2x` runtime `appraisedValue`), FIFO title reservation queue with 3-day pickup hold and in-app notification, librarian KPI dashboard, and Stretch enhancements (`ROLE_GUEST` portal, CSV export, live Open Library ISBN lookup helper).
- **Explicitly Out of Scope (`CON-002` .. `CON-004`):** (1) **Periodicals/serials and unpublished theses** (although ADU regulations allow borrowing 2 non-latest periodical issues for 2 days and restrict unpublished theses, LMS manages books only); (2) **Live ADU OBIS/OBS, LDAP/SSO, or RFID/smart-card hardware** (personal ID card verification `BR-016` is performed by the Librarian at the desk using `memberId`); (3) **External payment/banking APIs and external SMTP email/SMS gateways** (fines are settled at the desk as `PAID` / `WAIVED` ; hold-ready alerts are rendered in-app).



1.2 Stakeholders & User Personas

Full specification and circulation scenarios: `docs/02-inception/PERSONAS.md`. Every persona is anchored to real synthetic records in `data/seed/*.json`.

#	Persona & Role	Member Type (`memberType`)	Representative Seed Record (`data/seed/*.json`)	Confirmed Quota (BR-002)	Confirmed Period (BR-003)	Core Operational / Self-Service Goals
1	**Librarian / Admin** (`ROLE_LIBRARIAN`)	Staff Account (`ASSUMPTION-01`)	`syn-lib-0001` (`Enginiz Yıldırım`) in `librarians.json` *(colleagues: `syn-lib-0002`, `syn-lib-0003`)*	N/A	N/A	Fast desk checkout/return by `memberId` & `barcode`, catalog/copy CRUD, member management, fine settlement/waiver (`settledBy`), lost-copy assessment, KPI dashboard (`UC-03`..`UC-07`, `UC-09`, `UC-11`).

#	Persona & Role	Member Type (`memberType`)	Representative Seed Record (`data/seed/*.json`)	Confirmed Quota (`BR-002`)	Confirmed Period (`BR-003`)	Core Operational / Self-Service Goals
2	**Student Member** (`ROLE_MEMBER`)	`STUDENT`	`SYN-STU-0001` (`Tutkucan Ergül`, Chemistry, `LOAN-0099`) & `SYN-STU-0003` (`Hazrat Yorulmaz`, Computer Eng., `RES-0009` `READY` on `LMS-COPY-1599`)	**3 books**	**30 days**	Search course textbooks, check availability & `referenceOnly` badges, view own active/historical loans, cooldowns & fines, renew once online, join FIFO hold queue (`UC-01`, `UC-07`, `UC-08`, `UC-10`).
3	**Academic Staff Member** (`ROLE_MEMBER`)	`ACADEMIC_STAFF`	`SYN-STF-0003` (`Atasagun Öcalan`, Computer Eng., `RES-0007`) & `SYN-STF-0002` (`Baydu Arsoy`, Mathematics, `RES-0001`)	**7 books**	**30 days**	Borrow up to `7` research titles simultaneously (`AcademicStaffLoanPolicy`), inspect copy counts for course reading lists, renew non-reserved titles, reserve high-demand literature (`UC-01`, `UC-07`, `UC-08`, `UC-10`).
4	**Administrative Staff Member** (`ROLE_MEMBER`)	`ADMINISTRATIVE_STAFF`	`SYN-ADM-0001` (`Doğannur Güçlü`, Business Admin.) & `SYN-ADM-0004` (`Peren Erdoğan`, Computer Eng.)	**3 books**	**30 days**	Browse general & professional books, borrow up to `3` books in person with university staff ID card (`AdministrativeStaffLoanPolicy`), track due dates and renew online (`UC-01`, `UC-07`, `UC-10`).
5	**Guest Visitor** (`ROLE_GUEST`)	Unauthenticated	Anonymous browser session	**0 books**	N/A	Browse and filter the public catalog and inspect shelf availability without logging in, with zero access to member or loan data (`UC-01`, `FR-030`).

1.3 Supplementary Specification Overview (`FR`, `NFR`, `CON`, `DR`)

Full specification of all `32` Functional Requirements, `12` Non-Functional Requirements, `6` Constraints, and `7` Data Requirements: `docs/02-inception/REQUIREMENTS.md` and `data/README.md`.

Summary of Requirement Categories & Seed v2 Counts

Category	ID Range	Count	Breakdown & Scope Summary
Functional Requirements	`FR-001` .. `FR-032`	**32**	**19 Must Have** (`FR-001` .. `FR-019` across `M1` – `M6`), **10 Should Have** (`FR-020` .. `FR-029` across `S1` – `S4`), **3 Could Have** (`FR-030` .. `FR-032` across `C1` – `C3`).
Non-Functional Requirements	`NFR-001` .. `NFR-012`	**12**	Usability (`NFR-001` Stitch MCP UI-first, `NFR-002` desk barcode efficiency), Reliability (`NFR-003` `@Transactional` ACID, `NFR-004` 100% offline runtime), Performance targets (`NFR-005` `< 200 ms`, `NFR-006` `< 10 s` startup), Security/Privacy (`NFR-007` RBAC/validation, `NFR-008` BCrypt/no secrets, `NFR-009` synthetic PII), Maintainability/Testability/Portability (`NFR-010` .. `NFR-012`).
System Constraints	`CON-001` .. `CON-006`	**6**	English & mandatory header policy (`CON-001`), books-only scope (`CON-002`), no live ADU integration (`CON-003`), no payment/SMTP/SMS gateways (`CON-004`), `tools/set-author` shared-PC rule (`CON-005`), protected `main` & dual ADUZEM upload (`CON-006`).
Seed Data Requirements	`DR-001` .. `DR-007`	**7**	`books-openlibrary.json` (**`782` books**: `779` circulating, `3` `referenceOnly`), `copies.json` (**`1,706` copies**: `1,624` `AVAILABLE`, `59` `LOANED`, `15` `MAINTENANCE`, `5` `LOST`, `3` `RESERVED`), `members.json` (**`60` members**: `40` `STUDENT`, `10` `ACADEMIC_STAFF`, `10` `ADMINISTRATIVE_STAFF`; `55` `ACTIVE`, `5` `SUSPENDED`), `librarians.json` (**`3` librarians**), `loans.json` (**`164` loans**: `59` `ACTIVE`, `100` `RETURNED`, `5` `LOST`), `fines.json` (**`47` fines**: `26` `UNPAID` totaling `187.50 TRY`, `16` `PAID`, `5` `WAIVED`), `reservations.json` (**`22` reservations**: `19` `WAITING`, `3` `READY`).

Highlights of the Most Critical Functional & Non-Functional Requirements

ID(s)	Priority / Category	Condensed Requirement Statement	Linked `BR-*` / `UC-*`
`FR-001` .. `FR-005`	Must (`M1` Catalog & Copies)	Librarian CRUD on `Book` (`isbn13` checksum, 11 categories, `referenceOnly` flag for encyclopedias/dictionaries/catalogs), public/member search & detail view with `REFERENCE ONLY` badge, barcode registration (`LMS-COPY-*`), and GoF **State** machine (`BookCopyState`) over `AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`.	`BR-001`, `BR-014`, `BR-015`, `BR-017` · `UC-01`, `UC-03`

ID(s)	Priority / Category	Condensed Requirement Statement	Linked BR-* / UC-*
FR-009 .. FR-013	Must (M3 Checkout & Return)	Atomic @Transactional desk checkout verifying personal ID (memberId , no proxy), ACTIVE status, quota (7 / 3 / 3), zero overdue loans, zero UNPAID fines, 30-day post-renewal cooldown on isbn13 , and copy eligibility (!referenceOnly , AVAILABLE or matching READY hold); atomic return computing overdue fines and setting copy state.	BR-002 , BR-003 , BR-006 .. BR-008 , BR-011 , BR-013 , BR-014 , BR-016 , BR-017 · UC-05 , UC-06
FR-014 , FR-015	Must (M4 Overdue & Fines)	Zero-grace-period overdue calculation overdueDays = max(0, (effectiveEndDate - dueDate).days) (using lostReportedOn , returnDate , or currentDate) and uncapped daily fine amount = 0.25 * overdueDays TRY via FineCalculationStrategy .	BR-007 .. BR-010 , BR-015 · UC-06 , UC-09 , UC-11
FR-016 .. FR-019	Must (M5 Auth/Self-Service & M6 Seed Loader)	Local login/logout with startup-seeded BCrypt demo passwords (OPEN-05), endpoint RBAC (ROLE_LIBRARIAN , ROLE_MEMBER , ROLE_GUEST), member self-service portal + librarian inspection (FR-008 , FR-018), and startup JSON seed loader with date shifting (delta = today - 2026-10-22 , OPEN-04).	BR-001 , BR-002 , BR-006 , BR-010 , BR-013 , BR-017 , BR-018 · UC-02 , UC-04 , UC-10
FR-020 .. FR-023	Should (S1 FIFO Hold Queue)	Title-level reservation when 0 copies are AVAILABLE and !referenceOnly ; strict FIFO queuePosition = 1..n ; GoF **Observer** hold promotion on return (copy -> RESERVED , reservation -> READY , holdUntil = readyOn + 3 days , in-app notification); cancellation & 3-day expiry reallocation.	BR-011 .. BR-014 · UC-06 , UC-08 , UC-10
FR-024 .. FR-029	Should (S2 Renewal, S3 Fine/Lost, S4 Dashboard)	Single +30 -day renewal if not overdue and unreserved (FR-024); desk fine PAID and WAIVED settlement (FR-025 , FR-026); lost-copy petition (lostReportedOn freezes fine, 30-day window, 2.00 * appraisedValue , FR-027); dashboard KPI cards (FR-028) and overdue/unpaid report (FR-029).	BR-003 .. BR-005 , BR-007 .. BR-011 , BR-015 , BR-017 · UC-07 , UC-09 , UC-11
NFR-001 , 003 , 004 , 008 .. 011	Non-Functional Core	UI-first Stitch MCP prototypes in docs/03-design/ui/ before Thymeleaf views; @Transactional ACID integrity; 100% offline build & test execution; zero committed passwords/PII; 5-tier layered architecture with GoF/GRASP patterns and JUnit 5 + Python unittest suites.	Architecture & QA gates

1.4 Business Rules Overview (BR-001 .. BR-018)

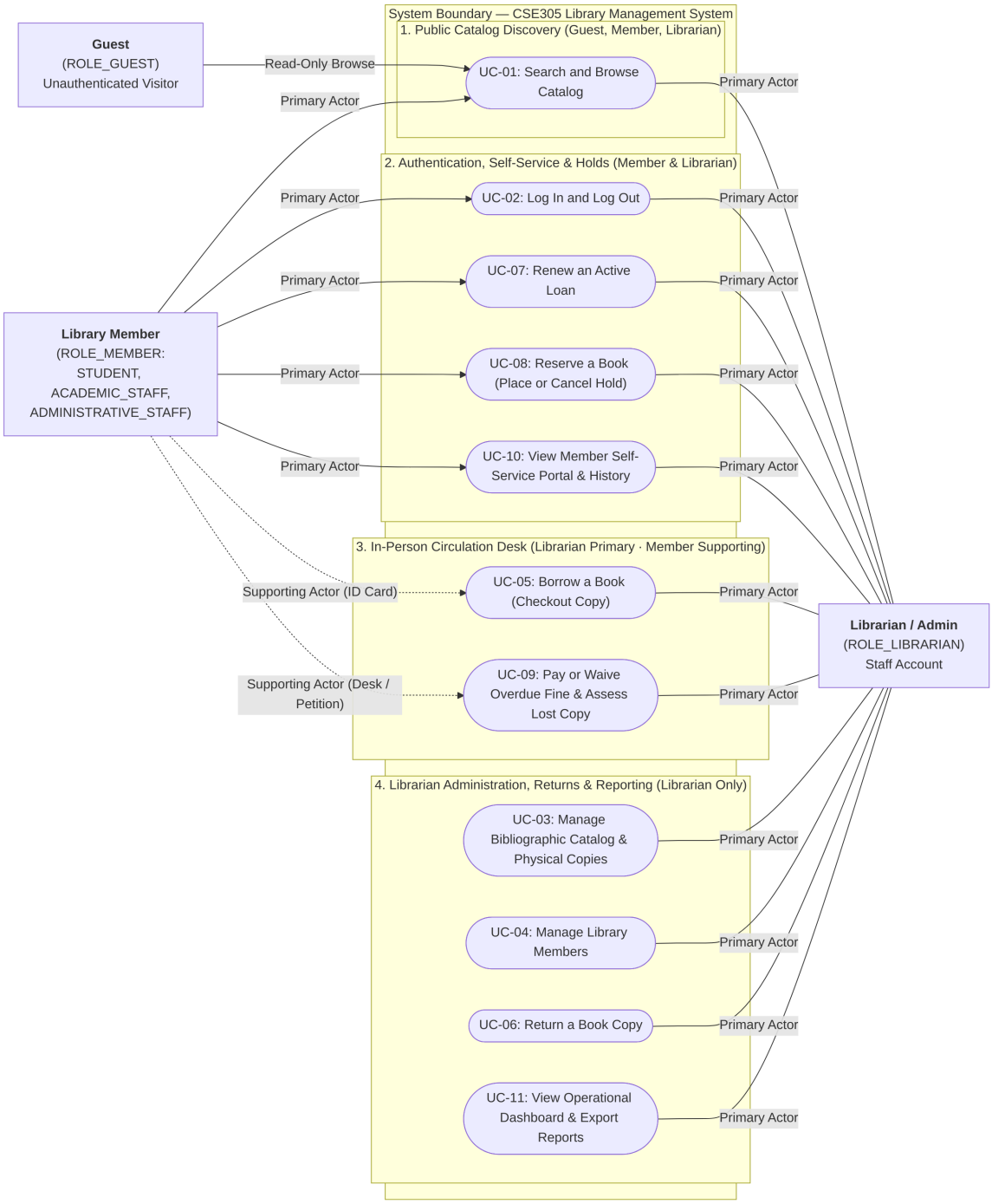
Full catalog of all 18 business rules, official ADU source citations (verified 2026-10-22), HW01 assumption disposition table (ASSUMPTION-01 .. 09), new assumptions (ASSUMPTION-10 .. 17), and exact BigDecimal / LocalDate formulas: docs/02-inception/BUSINESS_RULES.md .

Rule ID(s)	Rule Name	Summary of Rule & Deterministic Formula	Verification Status
BR-002 , BR-003	**Borrowing Quotas & 30-Day Loan Period**	Max simultaneous ACTIVE loans: **ACADEMIC_STAFF = 7** , **ADMINISTRATIVE_STAFF = 3** , **STUDENT = 3** . Standard loan duration is **30 calendar days** for all three member types (dueDate = checkoutDate + 30 days).	**CONFIRMED** *(ADU Official Rules, 2026-10-22)*
BR-004 , BR-005	**Single Renewal & Renewal Prohibitions**	An ACTIVE loan may be extended at most **once** (renewalCount: 0 -> 1 , dueDate = checkoutDate + 60 days), and **cannot** be renewed if currentDate > dueDate (overdue) or if any WAITING / READY reservation exists on isbn13 .	**CONFIRMED** (BR-004) + ASSUMPTION-11 (BR-005 : max 1 renewal of +30 days)
BR-006	**Post-Renewal 1-Month Re-Borrowing Cooldown**	When a renewed loan (renewalCount >= 1) is returned on returnDate , the same memberId cannot borrow any copy of the same isbn13 until returnDate + 30 calendar days (currentDate < returnDate + 30 days is rejected).	**CONFIRMED** *(1-month rule)* + ASSUMPTION-12 *(30-day mapping)*
BR-007 .. BR-010	**Overdue Days, 0.25 TRY/day Tariff (No Cap) & Settlement**	Zero grace period: overdueDays = max(0, (effectiveEndDate - dueDate).days) . Daily fine is **0.25 TRY (25 kuruş) per overdue day** with **no maximum cap** (amount = 0.25 * overdueDays). At most one Fine per loanId ; settled as PAID or WAIVED by a Librarian (settledOn , settledBy) after loan resolution.	**CONFIRMED** (0.25 TRY/day) + ASSUMPTION-10 *(no cap; 250 TRY cap removed)*
BR-011	**Borrowing & Renewal Block**	Any member holding **>= 1 overdue active loan** OR **>= 1 UNPAID fine (amount > 0.00 TRY)** OR status == SUSPENDED is blocked from borrowing new books or renewing existing loans.	**CONFIRMED** *(overdue & unpaid fine block)* + ASSUMPTION-06 *(SUSPENDED)*
BR-012 , BR-013	**FIFO Reservation Queue & 3-Day Pickup Hold**	Title-level hold (isbn13) allowed only when 0 copies are AVAILABLE (and member is ACTIVE , not already borrowing/queuing for isbn13). On copy return, copy -> RESERVED , queuePosition == 1 reservation -> READY (readyCopyBarcode , readyOn = returnDate , holdUntil = readyOn + 3 days), and an **in-app notification** is created.	**CONFIRMED** *(notification & 2–3 day hold)* + ASSUMPTION-13 *(FIFO, in-app, 3 days)*
BR-014	**Non-Circulating Reference Works (referenceOnly)**	**Encyclopedias, dictionaries, and catalogs** have referenceOnly = true and cannot be borrowed or reserved. *(Periodicals and unpublished theses are Out of Scope per CON-002).*	**CONFIRMED** *(ADU Official Rules)* + ASSUMPTION-14 *(referenceOnly boolean)*

Rule ID(s)	Rule Name	Summary of Rule & Deterministic Formula	Verification Status
`BR-015`	**Lost-Copy Petition, Fine Freeze & 2x Valuation Penalty**	Filing a lost-book petition (`lostReportedOn`) **immediately freezes daily overdue fine accrual on `lostReportedOn`** and opens a 30-day replacement window. If unreplaced after 30 days, `copy -> LOST` , `loan -> LOST` , and the Librarian enters the Valuation Commission's `appraisedValue` at runtime to assess **`compensationAmount = 2.00 * appraisedValue`** plus frozen overdue fine.	**`CONFIRMED`** *(freeze, 1-month window, 2x valuation)* + `ASSUMPTION-12` , `ASSUMPTION-15`
`BR-016` .. `BR-018`	**Personal ID Card, Copy State Invariant & Synthetic Auth**	Borrowers must present their own personal university ID card (`memberId` ; **no proxy borrowing**, `BR-016`). Every copy has a single valid state (`BR-017`). Seed JSON contains 100% synthetic PII and zero passwords; BCrypt demo passwords are initialized at startup (`BR-018`).	**`CONFIRMED`** (`BR-016`) + `ASSUMPTION-08` , `09` , `16` , `17`

1.5 Use Case Model Overview (`UC-01` .. `UC-11`)

Full specifications (including the **4 fully-dressed** use cases `UC-01` , `UC-05` , `UC-06` , `UC-09` with preconditions, postconditions, main success scenarios, and alternative/exception flows, plus the **7 brief** use cases): `docs/02-inception/USE_CASES.md` .



UC ID	Use Case Name	Primary Actor(s)	Detail Level in `USE_CASES.md`	Realized `FR-*` IDs	Enforced `BR-*` IDs	Implementing Stories
`UC-01`	**Search and Browse Catalog**	`Guest`, `Member`, `Librarian`	**Fully-Dressed**	`FR-002`, `FR-003`, `FR-030`	`BR-001`, `BR-014`, `BR-017`	`LMS-004` (S1), `LMS-018` (S3)
`UC-02`	**Log In and Log Out**	`Member`, `Librarian`	Brief	`FR-016`, `FR-017`	`BR-001`, `BR-018`	`LMS-003` (S1)
`UC-03`	**Manage Bibliographic Catalog & Physical Copies**	`Librarian`	Brief	`FR-001`, `FR-004`, `FR-005`, `FR-031`	`BR-001`, `BR-014`, `BR-017`	`LMS-005` (S1), `LMS-020` (S3)
`UC-04`	**Manage Library Members**	`Librarian`	Brief	`FR-006`, `FR-007`, `FR-008`	`BR-001`, `BR-002`, `BR-011`, `BR-018`	`LMS-006` (S2), `LMS-013` (S2)
`UC-05`	**Borrow a Book (Checkout Copy)**	`Librarian` *(Member in person)*	**Fully-Dressed**	`FR-005`, `FR-009`..`FR-011`, `FR-013`	`BR-002`, `BR-003`, `BR-006`, `BR-011`, `BR-013`, `BR-014`, `BR-016`, `BR-017`	`LMS-007` (S2), `LMS-017` (S3 hold pickup)
`UC-06`	**Return a Book Copy**	`Librarian`	**Fully-Dressed**	`FR-005`, `FR-012`..`FR-015`, `FR-022`	`BR-006`..`BR-010`, `BR-013`, `BR-017`	`LMS-010` (S2), `LMS-017` (S3 hold promo)
`UC-07`	**Renew an Active Loan**	`Member`, `Librarian`	Brief	`FR-024`	`BR-003`..`BR-006`, `BR-011`	`LMS-014` (S3)
`UC-08`	**Reserve a Book (Place or Cancel Hold)**	`Member`, `Librarian`	Brief	`FR-020`..`FR-023`	`BR-011`..`BR-014`	`LMS-017` (S3), `LMS-012` (S3)
`UC-09`	**Pay or Waive Overdue Fine & Assess Lost Copy**	`Librarian` *(Member)*	**Fully-Dressed**	`FR-005`, `FR-014`, `FR-015`, `FR-025`..`FR-027`	`BR-007`..`BR-011`, `BR-015`, `BR-017`	`LMS-009` (S2), `LMS-011` (S3), `LMS-016` (S3)
`UC-10`	**View Member Self-Service Portal & History**	`Member`, `Librarian`	Brief	`FR-008`, `FR-018`	`BR-001`, `BR-002`, `BR-006`, `BR-010`, `BR-013`	`LMS-013` (S2), `LMS-017` (S3 holds view)
`UC-11`	**View Operational Dashboard & Export Reports**	`Librarian`	Brief	`FR-014`, `FR-015`, `FR-028`, `FR-029`, `FR-032`	`BR-001`, `BR-007`, `BR-008`, `BR-011`	`LMS-015` (S3), `LMS-021` (S3), `LMS-019` (S3)

1.6 Domain Glossary & Conceptual Domain Model Bridge

Full ubiquitous language glossary (`20` entity/policy definitions + `8` enumeration tables): `docs/02-inception/DOMAIN_GLOSSARY.md`. Conceptual domain model (`7` classes, `50` attributes, `6` enums, `8` associations `A1`..`A8`, and narrative traceability `D1`..`D8`): `docs/01-proposal/HW01_Problem_Solution_DomainModel.md`.

Ubiquitous Term	Seed File (`data/seed/`)	Natural Key & Core Attributes	Key Domain Invariant
`Book`	`books-openlibrary.json` (`782`: `779` circulating, `3` `referenceOnly`)	`isbn13` (13-digit ISBN), `title`, `subtitle`, `authors`, `publisher`, `firstPublishYear`, `category` (11), `subjects`, `pageCount`, `coverId`, `coverUrl`, `openLibraryWorkKey`, `referenceOnly`	Owns `1..4` physical `BookCopy` items (`A1`). If `referenceOnly == true` (`3` books), copies cannot be loaned or reserved (`BR-014`).
`BookCopy`	`copies.json` (`1,706`: `1,624` `AVAILABLE`, `59` `LOANED`, `15` `MAINTENANCE`, `5` `LOST`, `3` `RESERVED`)	`barcode` (`LMS-COPY-1001`..), `isbn13` (FK), `status` (`AVAILABLE`, `LOANED`, `RESERVED`, `MAINTENANCE`, `LOST`)	Governed by `BookCopyState`; at most one `ACTIVE` loan (`59`) or one `READY` reservation (`3`) per copy at any moment (`BR-017`).
`Member`	`members.json` (`60`: `40` `STUDENT`, `10` `ACADEMIC_STAFF`, `10` `ADMINISTRATIVE_STAFF`; `55` `ACTIVE`, `5` `SUSPENDED`)	`memberId` (`SYN-STU-*`, `SYN-STF-*`, `SYN-ADM-*`), `firstName`, `lastName`, `email` (`@synthetic.lms.local`), `memberType`, `status` (`ACTIVE`, `SUSPENDED`), `department`, `registeredOn`	`memberId` represents the personal university ID card (`BR-016`). Quotas resolved via `LoanPolicyStrategy` (`7` / `3` / `3`, `BR-002`).
`Librarian`	`librarians.json` (`3` `ROLE_LIBRARIAN`)	`username` (`syn-lib-0001`..), `displayName`, `email`, `role` (`ROLE_LIBRARIAN`)	Zero password fields in JSON; BCrypt demo hashes seeded at startup (`BR-018`). Settles fines (`A5`).
`Loan`	`loans.json` (`164`: `59` `ACTIVE`, `100` `RETURNED`, `5` `LOST`)	`loanId` (`LOAN-0001`..), `memberId`, `copyBarcode`, `checkoutDate`, `dueDate`, `returnDate`, `status` (`ACTIVE`, `RETURNED`, `LOST`), `renewalCount` (`0..1`), `renewedOn`, `lostReportedOn`	`dueDate = checkoutDate + 30 * (1 + renewalCount)`. Renewed return triggers 30-day cooldown (`BR-006`). Lost petition freezes fine (`BR-015`).

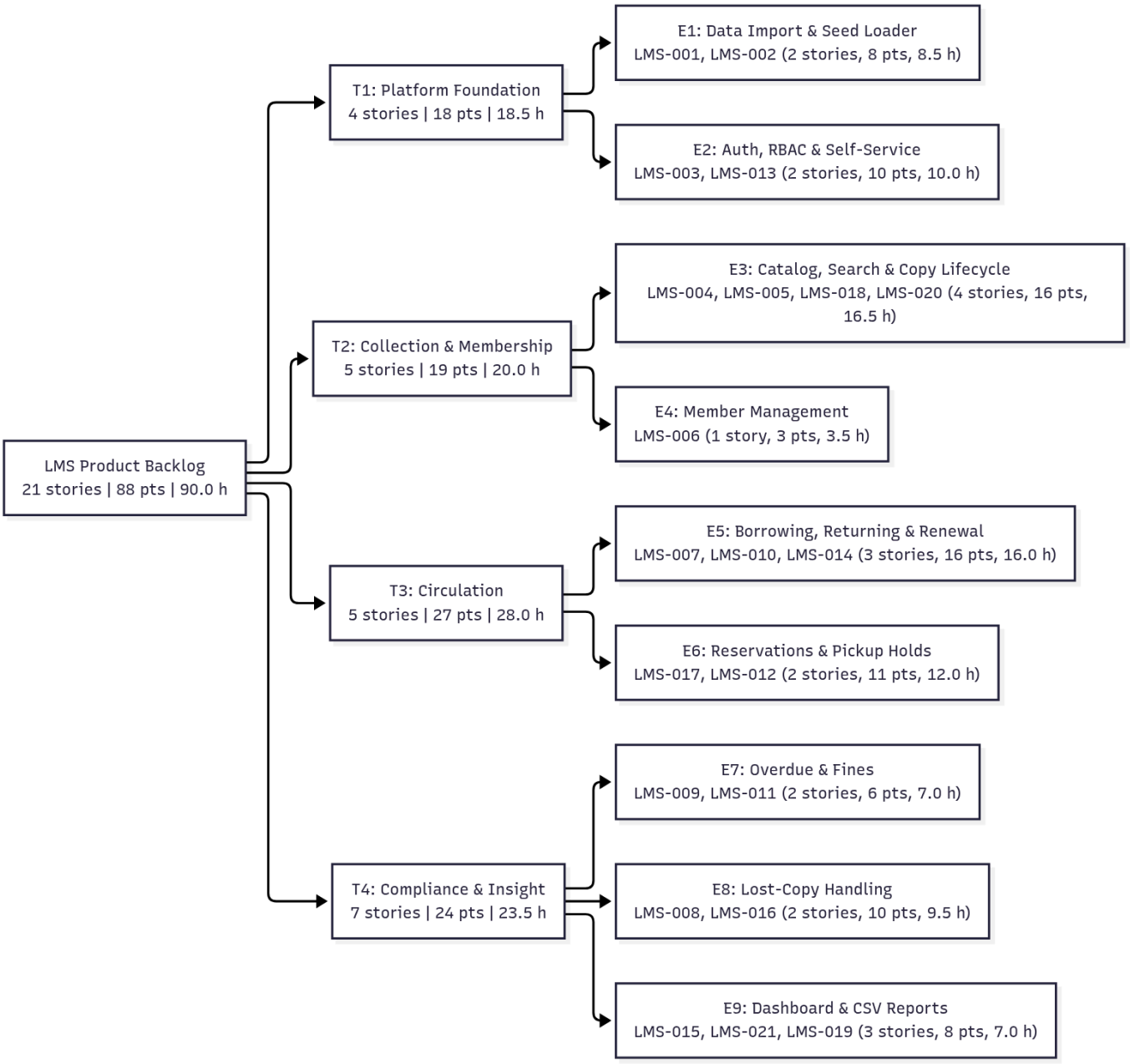
Ubiquitous Term	Seed File (data/seed/)	Natural Key & Core Attributes	Key Domain Invariant
Fine	fines.json (47 : 26 UNPAID [187.50 TRY], 16 PAID , 5 WAIVED)	fineId (FINE-0001 ..), loanId (unique FK), memberId , overdueDays , amount (0.25 * overdueDays), currency (TRY), status (UNPAID , PAID , WAIVED), assessedOn , settledOn , settledBy	Uncapped 0.25 TRY/day (BR-008 , BR-009). Any UNPAID fine blocks new checkouts and renewals (BR-011).
Reservation	reservations.json (22 : 19 WAITING , 3 READY)	reservationId (RES-0001 ..), isbn13 , memberId , queuePosition (1..n), status (WAITING , READY), createdOn , readyCopyBarcode , readyOn , holdUntil	Title-level FIFO queue when 0 copies are AVAILABLE (BR-012); READY hold binds RESERVED copy for 3 days (holdUntil = readyOn + 3 days , BR-013).

Part 2 — Agile Scrum Product Backlog & Iteration Plan

2.1 Themes & Epics (T1 – T4 , E1 – E9)

Full specification: docs/02-inception/THEMES_EPICS.md . The product is organized into **4 Themes** and **9 Epics** decomposing into **21 User Stories** (88 Story Points, 173.0 classic hours, 90.0 human hours) and **105 Technical Tasks**.

Theme	Epic	Goal	Stories	Points	Classic h	Human h	Assigned Story IDs (Sprint)
T1 Platform Foundation	—	Data and access foundations every feature relies on	**4**	**18**	**36**	**18.5**	—
↳ T1	**E1 Data import & seed loader**	Load seed v2 reproducibly with dates shifted to startup day (OPEN-04).	2	8	16	8.5	LMS-001 (S1), LMS-002 (S1)
↳ T1	**E2 Auth, RBAC & self-service**	Local login with BCrypt demo passwords (OPEN-05), RBAC, member self-service.	2	10	20	10.0	LMS-003 (S1), LMS-013 (S2)
T2 Collection & Membership	—	Who and what the library holds	**5**	**19**	**38**	**20.0**	—
↳ T2	**E3 Catalog, search & copy lifecycle**	Find books, maintain books/copies, enforce BookCopyState & referenceOnly .	4	16	32	16.5	LMS-004 (S1), LMS-005 (S1), LMS-018 (S3), LMS-020 (S3)
↳ T2	**E4 Member management**	Register, edit, and suspend synthetic members across 3 member types.	1	3	6	3.5	LMS-006 (S2)
T3 Circulation	—	Lending, returning, renewing, and holding books	**5**	**27**	**55**	**28.0**	—
↳ T3	**E5 Borrowing, returning & renewal**	Desk checkout, return, and single renewal under confirmed ADU rules.	3	16	32	16.0	LMS-007 (S2), LMS-010 (S2), LMS-014 (S3)
↳ T3	**E6 Reservations & pickup holds**	Title-level FIFO holds, in-app notification, 3-day pickup window, cancellation/expiry.	2	11	23	12.0	LMS-017 (S3), LMS-012 (S3)
T4 Compliance & Insight	—	Fines, lost copies, and management information	**7**	**24**	**44**	**23.5**	—
↳ T4	**E7 Overdue & fines**	Deterministic overdue days, uncapped 0.25 TRY/day fines, desk payment & waiver.	2	6	13	7.0	LMS-009 (S2), LMS-011 (S3)
↳ T4	**E8 Lost-copy handling**	Design spike (LMS-008) + lost-copy petition, 30-day window, 2x runtime appraisedValue .	2	10	16	9.5	LMS-008 (S1), LMS-016 (S3)
↳ T4	**E9 Dashboard & CSV reports**	Operational KPI cards (015a), overdue/unpaid report (015b), CSV export.	3	8	15	7.0	LMS-015 (S3), LMS-021 (S3), LMS-019 (S3)
Total	**9 Epics**	**Complete Product Backlog**	**21**	**88**	**173**	**90.0**	**11 Must, 7 Should, 3 Could**



2.2 User Stories, Estimation Model & Task Breakdown (`LMS-001` .. `LMS-021`)

Full Given/When/Then acceptance criteria for all 21 stories: `docs/02-inception/USER_STORIES.md` . Full breakdown of all 105 technical tasks: `docs/02-inception/TASKS.md` .

Estimation Methodology (Product Owner Approved)

1. **Fibonacci Story Points (`1, 2, 3, 5, 8` ; total `88 pts`):** Relative complexity, rule density, and cross-entity touchpoints (`Must = 48 pts` , `Should = 32 pts` , `Could = 8 pts`). Estimates re-calibrated in commit `df7eaa4` from the initial 20-story draft (`6b4265c` / `28c40d9`), see reconciliation note in `docs/process/BACKLOG_POINT_RECONCILIATION.md` ; the dashboard story was split into `LMS-015` (`015a` KPI cards, `2 pts`) and `LMS-021` (`015b` overdue/unpaid report table, `3 pts`) so the commitment line lands cleanly inside Sprint 3 net capacity.
2. **Classic Effort Hours (`Classic h` ; total `173.0 h` , Must `98.0 h`):** Traditional two-student implementation time without AI coding assistants (`~2.0 h/pt`).
3. **Human Effort Hours (`Human h` ; total `90.0 h` , Must `53.0 h` , Should `29.5 h` , Could `7.5 h`):** The actual human time of the two students—writing prompts, steering AI agents, reviewing generated code/docs, executing manual and automated tests, and reviewing Stitch UI screens. Agent runtime is **not** counted. Calculated from `~0.40 x` classic coding effort plus fixed human review/testing overhead, a `4.0 h` timeboxed design spike (`LMS-008`), and PO-approved safety reserves on the two most rule-dense stories (`LMS-005 = 7.5 h` , `LMS-007 = 9.0 h`). **All estimates are explicit planning assumptions to be re-calibrated with Sprint 1 actual hours.**
4. **Agent Sessions (`46` sessions):** Budgeted heavy AI coding/design sessions (`1..4` per story) tracked against weekly AI quota limits (`RISK-H03`).
5. **Task Breakdown (`105` tasks = `90.0` human h in `docs/02-inception/TASKS.md`):** Every story is decomposed into `1.0 – 4.0 h` technical tasks covering Stitch MCP UI prototyping (`NFR-001` , scheduled strictly before Thymeleaf view tasks),

domain/service/repository coding, and JUnit 5 TDD unit/integration tests (NFR-011), with an empty Actual human h column ready for Sprint 1–3 tracking.

Catalog of User Stories (LMS-001 .. LMS-021)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (Given / When / Then)	Traceability (FR / UC / BR)	Est. (Pts / Classic / Human h / Tasks)
LMS-001	**Domain model and persistence skeleton:** As a developer, I want the 7 JPA entities (Book , BookCopy , Member , Librarian , Loan , Fine , Reservation), 6 enums, and Spring Data repositories mapped to H2 so that all stories build on a tested persistence layer.	Schema starts cleanly on file H2 (./data/db/lms) and in-memory test H2 (jdbc:h2:mem:lms_test); Book (9780002615075) with BookCopy (LMS-COPY-1002), Member (SYN-STU-0001), and Loan (LOAN-0099) round-trip with all associations (A1 .. A8); unique barcode and Fine.loanId constraints reject duplicates.	FR-001 , FR-004 .. FR-006 , FR-011 , FR-015 , FR-019 · BR-010 , BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-002	**Seed loader with date shifting (OPEN-04):** As a librarian/evaluator, I want the app to load data/seed/*.json on startup when H2 is empty and shift all seed dates by delta = today - 2026-10-22 so that active, due-soon, and overdue scenarios stay fresh on any demo date.	Empty H2 loads exact seed v2 counts (782 books with 3 referenceOnly , 1,706 copies [1,624 AVAILABLE , 59 LOANED , 15 MAINTENANCE , 5 LOST , 3 RESERVED], 60 members [40/10/10 ; 55 ACTIVE , 5 SUSPENDED], 3 librarians, 164 loans [59 ACTIVE , 100 RETURNED , 5 LOST], 47 fines [26 UNPAID totaling 187.50 TRY , 16 PAID , 5 WAIVED], 22 reservations [19 WAITING , 3 READY]); non-empty H2 skips reload (idempotent); date shift preserves overdueDays (FINE-0022 on LOAN-0107) and holdUntil - readyOn == 3 days (RES-0009).	FR-019 , DR-001 .. DR-007 · BR-017 , BR-018	5 pts / 10 h / 5.0 h (5 tasks)
LMS-003	**Login/logout with BCrypt demo passwords (OPEN-05) and RBAC:** As a librarian or member, I want to log in with a startup-seeded BCrypt password and see only endpoints permitted for my role so that desk operations are protected.	data/seed/*.json contains zero password fields; startup seeds BCrypt hashes for syn-lib-0001 .. 0003 (ROLE_LIBRARIAN) and SYN-STU-* / SYN-STF-* / SYN-ADM-* (ROLE_MEMBER); ROLE_MEMBER gets HTTP 403 on librarian CRUD/desk URLs; invalid credentials show safe error; logout invalidates session + CSRF.	FR-016 , FR-017 , NFR-007 , NFR-008 · UC-02 · BR-001 , BR-018	5 pts / 10 h / 5.0 h (6 tasks)
LMS-004	**Catalog search and book detail:** As a member or librarian, I want to search books by partial title, author, or exact isbn13 , filter by category, availability, or referenceOnly , and inspect copy states on the detail page.	Query "Yaşar Kemal" or "9780002615075" returns *İnce Memed* with available/total copy counts; referenceOnly == true (9780824722876) shows REFERENCE ONLY — NOT FOR LOAN badge and disables borrow/reserve; detail page lists all copies (LMS-COPY-1002 .. 1004) and active hold queue count.	FR-002 , FR-003 · UC-01 · BR-001 , BR-014 , BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-005	**Book CRUD, copy registration and copy lifecycle:** As a librarian, I want to create/edit/archive Book records, register LMS-COPY-XXXX barcodes, and enforce BookCopyState transitions so that inventory stays accurate.	Invalid ISBN-13 checksum is rejected; valid book creation persists all fields including referenceOnly ; new copy accepts AVAILABLE / MAINTENANCE and rejects duplicate barcode; BookCopyState allows AVAILABLE <-> MAINTENANCE and blocks LOANED -> MAINTENANCE or checkout of LOANED / MAINTENANCE / LOST .	FR-001 , FR-004 , FR-005 · UC-03 · BR-014 , BR-015 , BR-017	8 pts / 16 h / 7.5 h (7 tasks)
LMS-008	**Design spike — compensation tracking & reservation terminal states:** As an architect, I want a timeboxed spike resolving HW01 §3.7 open design items before Sprint 3 coding starts so that LMS-016 and LMS-017 require zero schema rework.	Produces ADR in docs/03-design/ specifying (a) how 2.00 * appraisedValue lost-copy compensation is persisted and blocks borrowing under BR-011 , and (b) whether fulfilled/cancelled/expired reservations use terminal enum values or deletion; updates JPA skeleton with green tests.	FR-011 , FR-023 , FR-027 · UC-05 , UC-08 , UC-09 · BR-011 .. BR-013 , BR-015	2 pts / 4 h / 4.0 h (3 tasks)
LMS-006	**Member registration, profile update and suspension:** As a librarian, I want to register synthetic STUDENT , ACADEMIC_STAFF , and ADMINISTRATIVE_STAFF members, edit profiles, and toggle ACTIVE / SUSPENDED status.	Validates SYN-STU-* / SYN-STF-* / SYN-ADM-* ID and @synthetic.lms.local email (NFR-009); displays confirmed quota (7 / 3 / 3); toggling status to SUSPENDED immediately blocks new checkouts, renewals, and reservations (BR-011).	FR-006 , FR-007 · UC-04 · BR-001 , BR-002 , BR-011 , BR-018	3 pts / 6 h / 3.5 h (5 tasks)
LMS-007	**Checkout with eligibility, cooldown and referenceOnly enforcement:** As a librarian, I want to check out an eligible copy to an in-person member (memberId + copyBarcode) with full validation of ADU rules in one @Transactional step.	LoanPolicyStrategy enforces max 7 (ACADEMIC_STAFF) vs 3 (ADMINISTRATIVE_STAFF / STUDENT) and dueDate = checkoutDate + 30 days ; blocks SUSPENDED , overdue active loan (LOAN-0099), UNPAID fine (FINE-0022), referenceOnly == true (9780824722876), non-AVAILABLE copy, or active 30-day post-renewal cooldown (BR-006).	FR-009 .. FR-011 , FR-013 · UC-05 · BR-002 , BR-003 , BR-006 , BR-011 , BR-013 , BR-014 , BR-016 , BR-017	8 pts / 18 h / 9.0 h (7 tasks)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (Given / When / Then)	Traceability (FR / UC / BR)	Est. (Pts / Classic / Human h / Tasks)
LMS-009	**Overdue detection and uncapped fine calculation:** As a librarian, I want deterministic overdue days (effectiveEndDate > dueDate, zero grace) and uncapped 0.25 TRY/day fines (FineCalculationStrategy) so penalties match ADU rules.	effectiveEndDate == dueDate yields 0 overdue days and no fine; 1 day late yields 0.25 TRY UNPAID; 1,200 days late yields 300.00 TRY (no 250 TRY cap, BR-009); lostReportedOn freezes overdueDays as of petition date (LOAN-0103, BR-015).	FR-014, FR-015 · UC-06, UC-09, UC-11 · BR-007..BR-010, BR-015	3 pts / 6 h / 3.5 h (4 tasks)
LMS-010	**Return of a copy (AVAILABLE-only in Sprint 2):** As a librarian, I want to scan a returned copyBarcode to close its ACTIVE loan, finalize any overdue fine, record post-renewal cooldown anchor, and transition the copy to AVAILABLE.	Atomic @Transactional return sets returnDate = currentDate, status = RETURNED, transitions copy -> AVAILABLE (hold promotion to RESERVED is added in Sprint 3 by LMS-017), persists/updates UNPAID fine if returnDate > dueDate, and anchors 30-day cooldown if renewalCount >= 1.	FR-012..FR-015 · UC-06 · BR-006..BR-010, BR-017	3 pts / 6 h / 3.5 h (5 tasks)
LMS-013	**Member self-service portal & librarian member inspection:** As a member (or librarian inspecting a member), I want to view active loans, due dates, historical loans, 30-day cooldown timers, and fines (UNPAID / PAID / WAIVED).	Authenticated member (SYN-STU-0001) sees own active/historical loans, cooldown end dates, and fines (LOAN-0099) and cannot access another member's profile (403); Librarian (syn-lib-0001) inspects any member with quota (7 / 3 / 3) and block reason banner. *(Holds/notifications added in LMS-017).*	FR-008, FR-018 · UC-04, UC-10 · BR-001, BR-002, BR-006, BR-010, BR-011	5 pts / 10 h / 5.0 h (6 tasks)
LMS-017	**Reservation, FIFO queue, hold promotion & in-app notification:** As a member or librarian, I want to place a FIFO hold when 0 copies are AVAILABLE, and have a returned copy automatically held as RESERVED (READY) for 3 days with an in-app alert.	Rejects hold if >= 1 copy is AVAILABLE, referenceOnly == true, member blocked (BR-011), or already borrowing/queuing isbn13; assigns FIFO queuePosition = 1..n; on return (UC-06 Observer), promotes queuePosition == 1 to READY (readyCopyBarcode, holdUntil = readyOn + 3 days), sets copy -> RESERVED, creates in-app notification, and allows checkout of RESERVED copy only to queuePosition == 1 member.	FR-010..FR-012, FR-018, FR-020..FR-022 · UC-05, UC-06, UC-08, UC-10 · BR-011..BR-014, BR-017	8 pts / 16 h / 8.5 h (7 tasks)
LMS-011	**Fine payment (PAID) and waiver (WAIVED):** As a librarian, I want to record desk payment or administrative waiver of an UNPAID fine on a resolved loan so that the member's financial block is cleared with auditability.	Marks Fine.status = PAID or WAIVED with settledOn = currentDate and settledBy = librarian.username (syn-lib-0001); rejects settlement while loan is still ACTIVE; clears BR-011 block once all overdue loans and UNPAID balances reach zero.	FR-025, FR-026 · UC-09 · BR-010, BR-011	3 pts / 7 h / 3.5 h (5 tasks)
LMS-012	**Reservation cancellation and 3-day hold expiry:** As a member or librarian, I want to cancel a reservation and have unclaimed READY holds expire after 3 days (currentDate > holdUntil) so RESERVED copies are never locked forever.	Cancelling WAITING decrements trailing queuePositions; cancelling or expiring a READY hold promotes the next WAITING reservation (rebinding readyCopyBarcode and holdUntil = currentDate + 3 days) or returns copy.status to AVAILABLE if queue is empty.	FR-023 · UC-08 · BR-012, BR-013, BR-017	3 pts / 7 h / 3.5 h (5 tasks)
LMS-015	**Operational dashboard KPI summary cards (015a):** As a librarian, I want a dashboard with real-time KPI cards (titles, copies by state, active loans, due-soon 0..3 days, overdue loans, total UNPAID fine TRY) for instant health visibility.	Seeded dataset renders 782 titles (3 referenceOnly), 1,706 copies across all 5 states (1,624 AVAILABLE, 59 LOANED, 15 MAINTENANCE, 5 LOST, 3 RESERVED), 59 active loans, 12 due-soon (dueDate in [currentDate, currentDate + 3 days]), 15 overdue active loans, and 187.50 TRY across 26 UNPAID fines.	FR-028 · UC-11 · BR-001, BR-007, BR-008	2 pts / 4 h / 2.0 h (4 tasks)
LMS-014	**Loan renewal with reserved guard (Stretch):** As a member or librarian, I want to renew an eligible ACTIVE loan once (+30 days) when not overdue, not already renewed, and not reserved by another member.	renewalCount: 0 -> 1, renewedOn = currentDate, dueDate = checkoutDate + 60 days; rejects if renewalCount >= 1, currentDate > dueDate, any WAITING / READY reservation exists on isbn13, or borrower is blocked (BR-011).	FR-024 · UC-07 · BR-003..BR-006, BR-011	5 pts / 8 h / 3.5 h (5 tasks)
LMS-016	**Lost-copy process with runtime appraisedValue (Stretch):** As a librarian, I want to record a lost-book petition (lostReportedOn), freeze daily overdue fines, and after 30 days enter appraisedValue to bill 2.00 * appraisedValue.	Recording lostReportedOn freezes overdueDays; replacement within 30 days closes loan as RETURNED; after 30 days, Librarian enters appraisedValue (e.g., 450.00 TRY), system charges 900.00 TRY (2x) + frozen fine, sets copy -> LOST and loan -> LOST (returnDate = null).	FR-005, FR-014, FR-027 · UC-09 · BR-007, BR-011, BR-015, BR-017	8 pts / 12 h / 5.5 h (6 tasks)

ID	Story Title & User Story Statement (*As a … I want … so that …*)	Key Acceptance Criteria Highlights (`Given / When / Then`)	Traceability (`FR / UC / BR`)	Est. (`Pts / Classic / Human h / Tasks`)
`LMS-021`	**Actionable overdue loans & unpaid fines report (`015b`, *Stretch*):** As a librarian, I want a prioritized table of overdue active loans and `UNPAID` fines with direct links to return/settle so I can act on delinquencies.	Displays `loanId`, `memberId`, member name, `copyBarcode`, `title`, `dueDate`, `overdueDays` (descending), accrued `amount` (`0.25 TRY/day`), and quick action links to member profile, return desk, and fine settlement.	`FR-014`, `FR-015`, `FR-029` · `UC-11` · `BR-001`, `BR-007`, `BR-008`, `BR-011`	`3 pts` / `6 h` / `3.0 h` (`3` tasks)
`LMS-018`	**Public guest catalog portal (`C1`, *Stretch*):** As an unauthenticated guest (`ROLE_GUEST`), I want to browse/search the catalog and view copy availability in read-only mode without seeing any member or loan data.	Anonymous `GET /catalog` and `/catalog/{isbn13}` succeed (`200 OK`) showing availability and `referenceOnly` badges; zero member/loan PII is rendered; reserve/borrow buttons redirect to `/login`.	`FR-002`, `FR-003`, `FR-030` · `UC-01` · `BR-001`, `BR-014`	`2 pts` / `4 h` / `2.0 h` (`4` tasks)
`LMS-019`	**CSV export of overdue & inventory reports (`C3`, *Stretch*):** As a librarian, I want to download the overdue loans report and physical copy inventory as UTF-8 CSV files for offline auditing.	`ROLE_LIBRARIAN` downloads valid UTF-8 CSV (`text/csv`) with escaped fields and Turkish diacritics (`ğ, ş, ı, ö, ü, ç`) intact; non-librarians receive HTTP `403`.	`FR-032` · `UC-11` · `BR-001`	`3 pts` / `5 h` / `2.0 h` (`4` tasks)
`LMS-020`	**Open Library ISBN lookup helper (`C2`, *Stretch*):** As a librarian adding a book, I want an optional "Fetch by ISBN" helper (`OpenLibraryCatalogAdapter`) that pre-fills metadata and degrades gracefully when offline.	Online lookup populates title, authors, publisher, year, and cover URL; offline or timeout (`<= 3 s`) shows a non-blocking banner (*"Offline mode: please enter book metadata manually"*) keeping manual form entry 100% functional (`NFR-004`).	`FR-001`, `FR-031` · `UC-03` · `BR-001`	`3 pts` / `6 h` / `3.5 h` (`4` tasks)

2.3 Ordered Product Backlog & End-to-End Traceability

Authoritative source: `docs/02-inception/PRODUCT_BACKLOG.md`, validated by `python3 tools/agile/validate_backlog.py`.

Ordered Product Backlog (`LMS-001` .. `LMS-021`)

Order	ID	Epic	Story Title	MoSCoW	Pts	Classic h	Human h	Agent Sess.	Sprint	Depends On	Status
1	`LMS-001`	E1	Domain model and persistence skeleton	Must	3	6	3.5	2	S1	—	Planned (S1)
2	`LMS-002`	E1	Seed loader with date shifting	Must	5	10	5.0	3	S1	`LMS-001`	Planned (S1)
3	`LMS-003`	E2	Login/logout with BCrypt demo passwords and RBAC	Must	5	10	5.0	3	S1	`LMS-001`, `LMS-002`	Planned (S1)
4	`LMS-004`	E3	Catalog search and book detail	Must	3	6	3.5	2	S1	`LMS-002`, `LMS-003`	Planned (S1)
5	`LMS-005`	E3	Book CRUD, copy registration and copy lifecycle	Must	8	16	7.5	3	S1	`LMS-003`, `LMS-004`	Planned (S1)
6	`LMS-008`	E8	Design spike: compensation tracking and reservation terminal states	Must	2	4	4.0	1	S1	`LMS-001`	Planned (S1)
7	`LMS-006`	E4	Member registration, profile update and suspension	Must	3	6	3.5	2	S2	`LMS-003`	Planned (S2)
8	`LMS-007`	E5	Checkout with eligibility, cooldown and referenceOnly enforcement	Must	8	18	9.0	4	S2	`LMS-005`, `LMS-006`	Planned (S2)
9	`LMS-009`	E7	Overdue detection and uncapped fine calculation	Must	3	6	3.5	2	S2	`LMS-001`	Planned (S2)
10	`LMS-010`	E5	Return of a copy (copy to AVAILABLE only)	Must	3	6	3.5	2	S2	`LMS-007`, `LMS-009`	Planned (S2)
11	`LMS-013`	E2	Member self-service portal and librarian member inspection	Must	5	10	5.0	3	S2	`LMS-007`, `LMS-009`, `LMS-010`	Planned (S2)
12	`LMS-017`	E6	Reservation, FIFO queue, hold promotion and in-app notification	Should	8	16	8.5	4	S3	`LMS-010`, `LMS-008`	Planned (S3: **commit**)

Order	ID	Epic	Story Title	MoSCoW	Pts	Classic h	Human h	Agent Sess.	Sprint	Depends On	Status
13	LMS-011	E7	Fine payment and waiver	Should	3	7	3.5	2	S3	LMS-009, LMS-010	Planned (S3: **commit**)
14	LMS-012	E6	Reservation cancellation and hold expiry	Should	3	7	3.5	2	S3	LMS-017	Planned (S3: **commit**)
15	LMS-015	E9	Operational dashboard KPI summary cards (015a)	Should	2	4	2.0	1	S3	LMS-009, LMS-010	Planned (S3: **commit**)
16	LMS-014	E5	Loan renewal with reserved guard	Should	5	8	3.5	2	S3	LMS-007	Planned (S3: *stretch*)
17	LMS-016	E8	Lost-copy process with runtime appraised value	Should	8	12	5.5	3	S3	LMS-010, LMS-011, LMS-008	Planned (S3: *stretch*)
18	LMS-021	E9	Actionable overdue loans and unpaid fines report (015b)	Should	3	6	3.0	1	S3	LMS-009, LMS-010, LMS-015	Planned (S3: *stretch*)
19	LMS-018	E3	Public guest catalog portal	Could	2	4	2.0	1	S3	LMS-004	Planned (S3: *stretch*)
20	LMS-019	E9	CSV export of overdue and inventory reports	Could	3	5	2.0	1	S3	LMS-021	Planned (S3: *stretch*)
21	LMS-020	E3	Open Library ISBN lookup helper	Could	3	6	3.5	2	S3	LMS-005	Planned (S3: *stretch*)
Total	**21**	**E1–E9**	**11 Must (48 pts / 53.0 h), 7 Should (32 pts / 29.5 h), 3 Could (8 pts / 7.5 h)**	—	**88**	**173**	**90.0**	**46**	**S1–S3**	**0 forward deps**	**105 tasks (90.0 h)**

End-to-End Traceability Matrix (FR-001..FR-032 -> UC -> BR -> Story -> Sprint)

Every Must Have functional requirement (FR-001..FR-019) is assigned to a story in Sprint 1 or Sprint 2 (Must FR not covered in S1+S2: none).

FR-* ID	MoSCoW Priority	Realizing Use Case(s)	Enforced Business Rule(s)	Implementing Story	Target Sprint
FR-001, FR-004, FR-005	Must Have (M1)	UC-03, UC-05, UC-06, UC-09	BR-014, BR-015, BR-017	LMS-005	**S1**
FR-002, FR-003	Must Have (M1)	UC-01	BR-001, BR-014, BR-017	LMS-004	**S1**
FR-006, FR-007	Must Have (M2)	UC-04	BR-001, BR-002, BR-011, BR-018	LMS-006	**S2**
FR-008, FR-018	Must Have (M2, M5)	UC-04, UC-10	BR-001, BR-002, BR-006, BR-010, BR-011, BR-013	LMS-013	**S2**
FR-009, FR-010, FR-011, FR-013	Must Have (M3)	UC-05, UC-06	BR-002, BR-003, BR-006, BR-011, BR-013, BR-014, BR-016, BR-017	LMS-007	**S2**
FR-012	Must Have (M3)	UC-06	BR-007, BR-008, BR-013, BR-017	LMS-010	**S2**
FR-014, FR-015	Must Have (M4)	UC-06, UC-09, UC-11	BR-007, BR-008, BR-009, BR-010, BR-015	LMS-009	**S2**
FR-016, FR-017	Must Have (M5)	UC-02	BR-001, BR-018	LMS-003	**S1**
FR-019	Must Have (M6)	System Startup	BR-017, BR-018	LMS-002 (on LMS-001)	**S1**
FR-020, FR-021, FR-022	Should Have (S1)	UC-06, UC-08, UC-10	BR-011, BR-012, BR-013, BR-014	LMS-017	**S3 (Commit)**
FR-023	Should Have (S1)	UC-08	BR-012, BR-013	LMS-012	**S3 (Commit)**

FR-* ID	MoSCoW Priority	Realizing Use Case(s)	Enforced Business Rule(s)	Implementing Story	Target Sprint
FR-024	Should Have (S2)	UC-07	BR-003, BR-004, BR-005, BR-011	LMS-014	**S3 (Stretch)**
FR-025, FR-026	Should Have (S3)	UC-09	BR-010, BR-011	LMS-011	**S3 (Commit)**
FR-027	Should / Could (S3)	UC-09	BR-005, BR-015, BR-017	LMS-016 (on LMS-008)	**S3 (Stretch)**
FR-028	Should Have (S4)	UC-11	BR-001, BR-007, BR-008	LMS-015 (015a)	**S3 (Commit)**
FR-029	Should Have (S4)	UC-11	BR-007, BR-008, BR-011	LMS-021 (015b)	**S3 (Stretch)**
FR-030	Could Have (C1)	UC-01	BR-001, BR-014	LMS-018	**S3 (Stretch)**
FR-031	Could Have (C2)	UC-03	BR-001	LMS-020	**S3 (Stretch)**
FR-032	Could Have (C3)	UC-11	BR-001	LMS-019	**S3 (Stretch)**

2.4 Three-Sprint Iteration Plan, Capacity Model & Commitment Line

Authoritative source: `docs/02-inception/SPRINT_PLAN.md`.

Sprint Schedule & Capacity Model

- **Capacity Unit (`Human h`):** Two students' human effort (`8 h/person/week` in normal weeks, reduced to `3 h/person/week` during midterm weeks `Nov 17 – Nov 28`, prorated per calendar day).
- **Fixed Deductions per Sprint:** Scrum ceremonies = `4.0 h` (`2 people x 2 h`); Homework engineering/design documentation = `8.0 h` in Sprint 1 (`HW03`: Use Case diagram `1.5 h`, System Sequence Diagrams `2.0 h`, Class Diagram `2.0 h`, TDD/Git/Maven plan `1.0 h`, Stitch UI review `1.5 h`), `4.0 h` in Sprint 2 (`HW04`), and `7.0 h` in Sprint 3 (`HW05`: user guide, test report, final retrospective); Contingency buffer = **`10%` of gross capacity** (never reduced).
- **Sign Convention for `Planned - Net`:** Positive (`+`) = overrun / over net capacity (`over`); Negative (`-`) = unused net capacity (`slack`).

Sprint	Calendar Window	Course Deliverable	Gross (h)	Ceremonies (h)	HW Docs (h)	10% Buffer (h)	Net Story Capacity (h)	Planned Stories (h)	Must (h)	Should (h)	Could (h)	Planned - Net
Sprint 1	Nov 11 – Dec 08 (28 d, incl. midterms Nov 17–28)	`HW03` (Elaboration + S1)	`46.9`	`4.0`	`8.0`	`4.7`	**`30.2`**	`28.5` (`26 pts`)	`28.5`	`0.0`	`0.0`	`-1.7 h (slack)`
Sprint 2	Dec 09 – Dec 22 (14 d)	`HW04` (Construction + S2)	`32.0`	`4.0`	`4.0`	`3.2`	**`20.8`**	`24.5` (`22 pts`)	`24.5`	`0.0`	`0.0`	`+3.7 h (over)`
Sprint 3	Dec 23 – Jan 05 (14 d)	`HW05` (Transition + MVP)	`32.0`	`4.0`	`7.0`	`3.2`	**`17.8`**	`17.5` committed (`+19.5` stretch)	`0.0`	`29.5`	`7.5`	`-0.3 h (slack, committed) / +19.2 h (over w/ stretch)`
Total	**Nov 11 – Jan 05**	**`HW03 – HW05`**	**`110.9`**	**`12.0`**	**`19.0`**	**`11.1`**	**`68.8`**	**`70.5` committed (`90.0` all)**	**`53.0`**	**`29.5`**	**`7.5`**	**S1+S2 Must: `+2.0 h` (over)**

Must Gap Analysis (`+2.0 h`, `4.0%` of S1+S2 Net Capacity — Accepted Risk `RISK-H01`)

All `11` Must stories require **`53.0` human hours** (`28.5 h` in S1 + `24.5 h` in S2) against a combined S1+S2 net story capacity of **`51.0` hours** (`30.2 + 20.8 h`), leaving a **known gap of `+2.0 h` (`4.0%` of net)**. Rather than artificially lowering the estimates of complex stories (`LMS-005 7.5 h`, `LMS-007 9.0 h`) or cutting the 10% buffer, the Product Owner approved three governance rules:

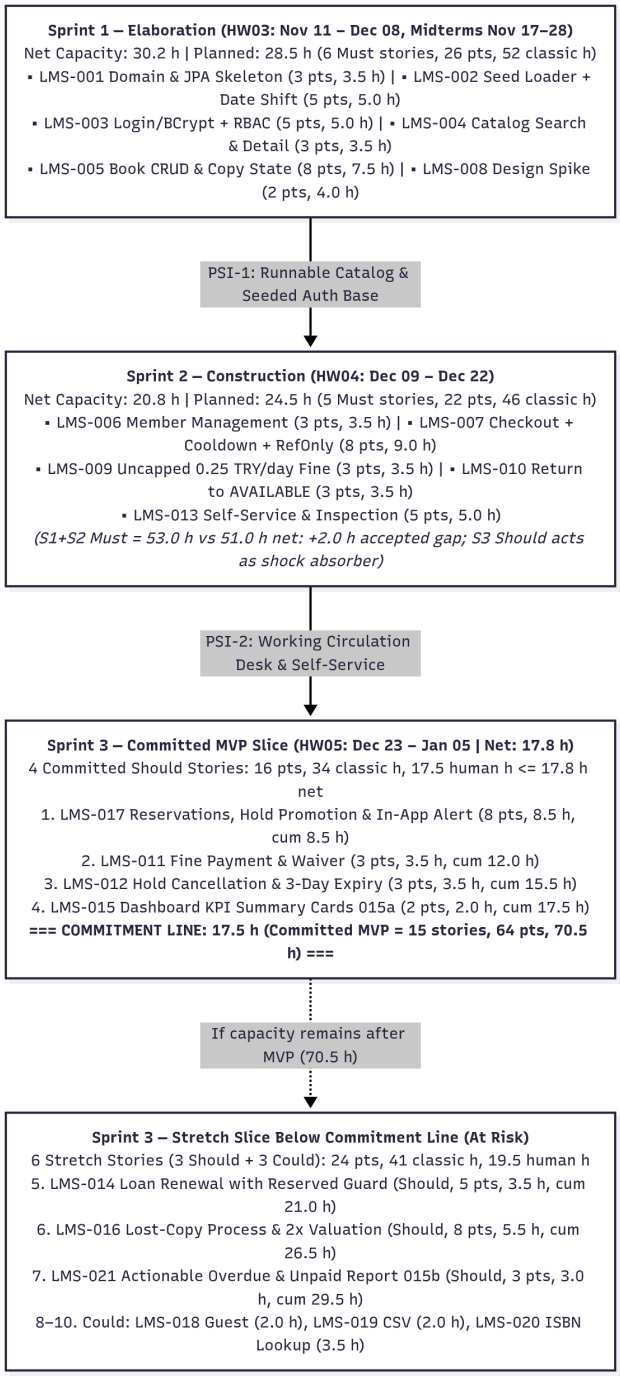
1. **Buffer Preservation:** The 10% buffer (`4.7 h` in S1, `3.2 h` in S2, `3.2 h` in S3) is **not** reduced.
2. **Sprint 3 Should Stories as Shock Absorber:** If Must stories overrun in Sprint 2, the overrun is absorbed by deferring Sprint 3 Should stories from the bottom of the commitment line upward.

3. **Quantitative Re-Plan Trigger:** If at the end of Sprint 1 the actual ratio `actual_human_h / classic_h` exceeds **`0.45`**, or if the remaining Sprint 2 Must estimate exceeds Sprint 2 net capacity (`20.8 h`) by more than **`10%`**, the Sprint 2 plan is re-baselined and any scope decision is brought to the Product Owner.

Sprint 3 Value Ordering & Commitment Line (**`17.5 h <= 17.8 h`** Net Capacity)

Because total Sprint 3 Should (`29.5 h`) + Could (`7.5 h`) = `37.0 h` exceeds Sprint 3 net capacity (`17.8 h`), the Product Owner ordered the Sprint 3 stories by domain value and state-machine consistency and drew an explicit **commitment line at `17.5 h`**:

S3 Rank	Story ID & Title	MoSCoW	Pts	Classic h	Human h	Cumulative Human h	Commitment Status & Domain Rationale
1	`LMS-017` Reservation, FIFO queue, hold promotion & in-app notification	Should (`S1`)	8	16	`8.5`	`8.5 h`	**COMMIT** — Core FIFO hold queue, `RESERVED` copy promotion on return, and 3-day pickup notification (`FR-020`..`FR-022`, `BR-012`, `BR-013`).
2	`LMS-011` Fine payment and waiver	Should (`S3`)	3	7	`3.5`	`12.0 h`	**COMMIT** — Desk `PAID`/`WAIVED` settlement (`FR-025`, `FR-026`, `BR-010`). Required so members with fines accrued in S2 can clear their `BR-011` borrowing block.
3	`LMS-012` Reservation cancellation and hold expiry	Should (`S1`)	3	7	`3.5`	`15.5 h`	**COMMIT** — Cancellation & 3-day hold expiry reallocation (`FR-023`). Required for reservation state-machine completeness so an unclaimed `READY` hold never locks a `RESERVED` copy forever.
4	`LMS-015` Operational dashboard KPI summary cards (`015a`)	Should (`S4`)	2	4	`2.0`	**`17.5 h <= 17.8 h`**	**COMMIT (Commitment Line)** — Delivers real-time librarian KPI cards (`FR-028`: titles, copies by state, active/due-soon/overdue loans, unpaid fine total).
5	`LMS-014` Loan renewal with reserved guard	Should (`S2`)	5	8	`3.5`	`21.0 h`	*STRETCH (below line)* — Single `+30`-day self-service/desk renewal (`FR-024`, `BR-004`, `BR-005`).
6	`LMS-016` Lost-copy process with runtime `appraisedValue`	Should (`S3`)	8	12	`5.5`	`26.5 h`	*STRETCH (below line)* — Lost-book petition fine freeze, 30-day window, and `2.00 * appraisedValue` runtime charge (`FR-027`, `BR-015`).
7	`LMS-021` Actionable overdue loans & unpaid fines report (`015b`)	Should (`S4`)	3	6	`3.0`	`29.5 h`	*STRETCH (below line)* — Detailed overdue/unpaid report table with quick desk links (`FR-029`).
8–10	`LMS-018` Guest portal (`2.0 h`), `LMS-019` CSV export (`2.0 h`), `LMS-020` ISBN lookup (`3.5 h`)	Could (`C1`–`C3`)	8	15	`7.5`	`37.0 h`	*STRETCH (below line)* — Optional enhancements (`FR-030`..`FR-032`) attempted only if all committed and Should stretch items finish early.



Cross-Sprint Dependency Fixes, DoR, DoD & Risk Register

- **Dependency Hygiene (Verified by `validate_backlog.py` -> `OVERALL: PASS`):** No story depends on a story in a later sprint. Specifically: (1) `LMS-010` (S2 Return) transitions returned copies to `AVAILABLE` only, while `RESERVED` hold promotion is added in S3 by `LMS-017`; (2) `LMS-013` (S2 Self-Service) displays active/historical loans, cooldowns, and fines only, while holds and in-app notifications are added in S3 by `LMS-017`; (3) `LMS-014` (S3 Renewal) checks active reservations via a repository query; and (4) `LMS-008` (S1 Spike) resolves compensation and reservation terminal-state design before S3.
- **Definition of Ready (DoR):** Standard user story format + testable Given/When/Then criteria with `FR`/`UC`/`BR` tags; Fibonacci points, classic/human hours, and agent sessions estimated; zero forward dependencies; Stitch MCP UI task scheduled before any Thymeleaf task; required seed test data identified in `data/seed/*.json`.
- **Definition of Done (DoD):** `./mvnw -q test` (or `tools/mvnw-vm.sh -q test` on the VM) and `python3 -m unittest discover -s tools/data/tests` pass 100% offline; mandatory CSE305 header on all first-party files; `docs/process/ai-usage-log.md` updated with exact model selector name and `"Pending human review"` until explicitly confirmed by a team member; cross-model review completed; UI matches Stitch prototype; merged to `develop` with sprint review/retrospective notes.

Risk ID	Risk Description	Impact	Mitigation & Trigger
`RISK-H01`	**Must gap of `+2.0 h` in S1+S2** (`53.0 h` Must vs `51.0 h` net)	Sprint 2 schedule pressure	Accepted by PO; 10% buffer kept intact; S3 Should stories absorb overrun; re-plan if S1 `human/classic > 0.45` or S2 Must exceeds net by `> 10%`.

Risk ID	Risk Description	Impact	Mitigation & Trigger
`RISK-H02`	**Sprint 3 Stretch stories (`19.5 h`) exceed S3 net (`17.8 h`)**	Stretch items (`LMS-014` , `016` , `021` , `018` .. `020`) not reached	Explicit commitment line at `17.5 h <= 17.8 h` ; MVP (`70.5 h`) requires only the 4 committed Should stories above the line.
`RISK-H03`	**Weekly AI quota exhaustion** across `46` budgeted agent sessions	Interruption mid-story	Commit & push after every file/step; keep `docs/process/*_PROGRESS.md` updated; budget agent sessions per week.
`RISK-H04`	**Human-hour conversion ratio (`~0.40x`) is an assumption**	Velocity variance	Track actual human hours in `TASKS.md` during Sprint 1 and re-calibrate Sprint 2–3 estimates at Sprint 1 Review.
`RISK-H05` .. `H08`	**Midterm capacity drop (`Nov 17–28`), VM proxy Maven wrapper, HW01 open design items, dual ADUZEM upload**	Schedule/build/submission issues	Midterms prorated (`3 h/wk`); `tools/mvnw-vm.sh` isolated from standard `./mvnw` ; `LMS-008` spike in S1; mandatory dual-upload check.

2.5 Potentially Shippable Increments (PSI) & MVP Definition

Authoritative source: `docs/02-inception/PSI_AND_MVP.md` .

Increment	Included Stories	Committed Effort	End-to-End Demonstrable Capability (PSI / MVP)
Sprint 1 PSI (`HW03`)	`LMS-001` .. `LMS-005` , `LMS-008` (`6` Must stories)	`26 pts` / `52` classic h / **`28.5` human h**	Application boots offline, loads seed v2 (`782` books, `1,706` copies) with shifted dates (`OPEN-04`), authenticates Librarians and Members via BCrypt demo passwords (`OPEN-05`) and RBAC, supports catalog search/filtering and `referenceOnly` badges, enforces `BookCopyState` on Book/Copy CRUD, and documents the `LMS-008` ADR.
Sprint 2 PSI (`HW04`)	+ `LMS-006` , `LMS-007` , `LMS-009` , `LMS-010` , `LMS-013` (`5` Must stories)	`22 pts` / `46` classic h / **`24.5` human h** *(Cumulative Must: `48 pts` / `53.0 h`)*	Complete circulation desk without holds: Librarian registers/suspends members, executes atomic checkouts (`7` / `3` / `3` quotas, `30` -day period, `referenceOnly` & block checks, 30-day post-renewal cooldown) and returns (`AVAILABLE`), computes uncapped `0.25 TRY/day` fines, and Members inspect active/historical loans, cooldowns, and fines.
Sprint 3 PSI — Committed MVP (`HW05`)	+ `LMS-017` , `LMS-011` , `LMS-012` , `LMS-015` (`4` Committed Should stories above the `17.5 h` line)	`16 pts` / `34` classic h / **`17.5` human h** ***(Total Committed MVP: `15` stories, `64 pts` , `132` classic h, `70.5` human h)***	Adds title-level FIFO reservations, automatic `RESERVED` hold promotion on return with 3-day pickup window and in-app notification (`LMS-017`), desk fine `PAID` / `WAIVED` settlement (`LMS-011`), reservation cancellation and 3-day hold expiry reallocation (`LMS-012`), and operational dashboard KPI summary cards (`LMS-015`).
Sprint 3 Stretch Slice *(Below Line, At Risk)*	`LMS-014` , `LMS-016` , `LMS-021` , `LMS-018` .. `LMS-020` (`3` Should + `3` Could stories)	`24 pts` / `41` classic h / **`19.5` human h** *(Full backlog if all stretch completed: `21` stories, `88 pts` , `90.0 h`)*	Self-service/desk renewal (`LMS-014`), lost-copy petition & `2x` runtime `appraisedValue` assessment (`LMS-016`), actionable overdue/unpaid report table (`LMS-021`), public guest catalog (`LMS-018`), CSV export (`LMS-019`), and live Open Library ISBN lookup helper (`LMS-020`).

Revision History

Version	Date	Summary
`1.0`	2026-11-07	First consolidated HW02 submission document: Part 1 consolidates the UP Inception Overview (`VISION_AND_CONTEXT.md` , `PERSONAS.md` , `REQUIREMENTS.md` , `BUSINESS_RULES.md` , `USE_CASES.md` , `DOMAIN_GLOSSARY.md`) with embedded System Context and Use Case diagrams; Part 2 presents the Product Owner-approved Agile Scrum Product Backlog (`21` stories `LMS-001` .. `LMS-021` , `88` story points, `173.0` classic hours, `90.0` human hours, `105` tasks), 3-sprint capacity model, accepted `+2.0 h` Must gap governance (`RISK-H01`), Sprint 3 commitment line (`17.5 h <= 17.8 h` net), Themes/Epics and Sprint Roadmap diagrams, dependency proof (`OVERALL: PASS`), and PSI/MVP definition (`64 pts / 70.5 h` committed).